FACULTY OF COMPUTING

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SECP3843 – SPECIAL TOPIC IN DATA ENGINEERING

SECTION 1

TUTORIAL MICROSOFT AZURE

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1.0 PROJECT OVERVIEW

Nowadays, practically all businesses produce massive amounts of data simply from their daily operations, including customer records, sales transactions and product details. The challenge is that the majority of the data is distributed across multiple systems and saved in various formats, making it difficult to analyze and use for decision making. Without a well-designed data pipeline, the process of generating meaningful insights becomes inefficient and impractical.

This project aims to create an end-to-end data engineering pipeline using Microsoft Azure to transform raw data into useful business insights. The pipeline will show how cloud technology can automate data processing and visualization. This project's dataset contains sales-related information such as client names, items and transactions. This type of dataset is frequently utilized in real-world business scenarios to analyze sales performance and customer behaviour.

1.1 PROBLEM STATEMENT

Organizations often face challenges in analyzing raw data due to the lack of an automated system for data processing. Data stored in raw formats can't be utilized directly for reporting or decision making as it must be cleaned and transformed before it can be used. Performing these tasks manually is time consuming and increases the risk of errors. Therefore, there is a need for a system that can efficiently extract, transform and visualize data.

1.2 OBJECTIVE

The objective of this project are:

1. To extract data from a source and load it into the cloud.
2. To transform the raw data into a clean and structured format.
3. To store processed data for analysis.
4. To create an interactive dashboard for data visualization.
5. To automate the data pipeline using Azure services.

1.3 SCOPE

The scope of this project covers the full data pipeline process using Microsoft Azure. The key components included in this project:

1. Data Ingestion
   Data is extracted and loaded into the cloud using Azure Data Factory.
2. Data Storage
   Data is stored in Azure Data Lake Storage and organized into three layers which are Bronze layer, Silver layer and Gold layer.
3. Data Transformation
   Data is cleaned, structured and processed using Azure Databricks.

4. Data Visualization
   Processed data is visualized using Microsoft Power BI through interactive dashboards.

1.4 BUSINESS / INDUSTRY RELEVANCE

This project represents a real-world scenario in which organizations need to examine sales and customer data to make decisions. The capacity to handle and visualize data efficiently is critical in businesses like retail, e-commerce and banking. Organizations that use cloud-based technologies can enhance scalability, reduce manual work and receive faster insights.

2.0 DESCRIPTION ON THE SOLUTION

The solution is an end-to-end data pipeline built on Microsoft Azure that automates data ingestion, storage, transformation and visualization. The pipeline guarantees that raw data is turned into useful insights with minimal manual effort.

Step 1: Data Ingestion using Azure Data Factory
Azure Data Factory is in charge of extracting data from the source and loading it into the cloud. It automates data flow through a number of pipelines and operations, making the transfer more efficient and consistent.

Step 2: Data Storage in Azure Data Lake
The ingested data is stored in Azure Data Lake Storage, which acts as a centralized storage system. Data can be stored in different layers, namely the Bronze layer, Silver layer and Gold layer to organize and manage it effectively.

Step 3: Data Transformation using Databricks
After the data is stored, it goes through a transformation process using Azure Databricks. The process includes cleaning, formatting, filtering and combining data from different sources. The data is progressively transformed from the Bronze layer to the Silver and Gold layers to ensure that the data is consistent, accurate and ready to be analyzed.

Step 4: Data Visualization using Power BI
The processed data is visualized using Microsoft Power BI, which generates dashboards and reports. These dashboards provide essential information such as trends, performance and comparisons to help users make informed decisions.

3.0 TECHNOLOGY USE

This project uses a variety of Microsoft Azure services to create a fully automated data pipeline. Each technology contributes differently to ensuring that data flows smoothly from source to final visualization.

Microsoft Azure
Microsoft Azure is the primary cloud platform used in this project, providing the foundation and

environment required to develop the entire pipeline. It enables many services, such as data ingestion, storage and visualization to work together smoothly on a single platform.

Azure Data Factory
Azure Data Factory serves as the project's data integration and management tool. It is in charge of extracting data from the source and loading it into a cloud storage system.

Azure Data Lake Storage
Azure Data Lake Storage serves as the project's central storage infrastructure. It organizes raw and processed data into layers such as Bronze, Silver and Gold. The Bronze layer contains raw data, the Silver layer contains cleaned and transformed data and the Gold layer contains data that is ready for analysis.

Azure Databricks
Azure Databricks is a platform for data translation and processing. It offers a platform for performing data processing activities such as cleaning, filtering and combining the datasets. In this project, Databricks helps transform raw data into structured and analysis-ready data by moving it from the Bronze layer to the Silver and Gold layers.

Microsoft Power BI
Microsoft Power BI is a tool for data visualization and reporting. It connects to the processed data, allowing users to generate interactive dashboards and reports. These dashboards display important insights such as trends, comparisons and performance.

4.0 DATA PIPELINE ARCHITECTURE

The chosen architecture for this project solution is the Medallion Architecture. This architecture stores and organizes data in a data lake, and then it will undergo a three-layer processing cycle consisting of the Bronze, Silver, and Gold layers before being analyzed and visualized.

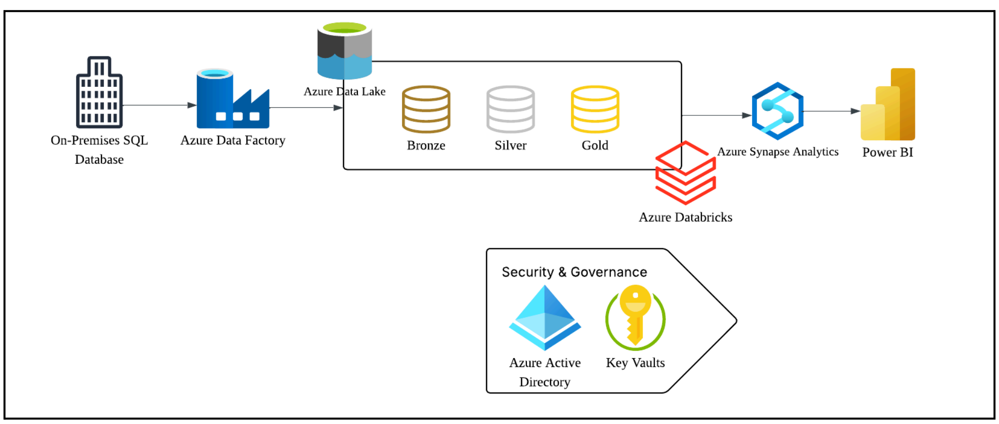

Figure 1: Medallion Architecture

4.1 DATA INGESTION

The first step of the architecture is Data Ingestion, where Azure Data Factory utilizes a Copy Activity to ingest raw data from sources such as On-Premises SQL Databases. This data is landed into the Bronze Layer of the Azure Data Lake Storage (ADLS) Gen2. In this layer, the data is stored in its original, raw format (such as Parquet or CSV) to ensure a complete 'source of truth' is maintained before any transformations occur.

4.2 DATA TRANSFORMATION

Once the raw data is landed, the transformation process begins. We utilize Azure Databricks to read the Parquet files from the Bronze layer and also read the delta files from the Silver layer.

a) **Automation cleansing**: Instead of going through the dataset row-by-row, we implemented a Python loop to automate the transformation across all tables (e.g., Address, Customer, Product).
b) **Standardization**: After we identified the raw data's date format complexity, we applied spark functions to standardize the format into yyyy-MM-dd across the dataset.
c) **Data curation**: The process where it moves data from silver to gold is also involved during the data transformation in which we use databricks to rename and structure back the data naming to make the data "business-ready".
d) **Delta Lake Implementation**: All transformed data was saved in Delta format. Unlike standard Parquet, Delta allows for better schema enforcement and data reliability moving towards the final visualization stage.

4.3 SERVING LAYER & VISUALIZATION

In the final stages, Azure Synapse Analytics and Power BI are involved to make the data presentable.

a) **Data warehousing**: The curated data from Gold layer are loaded into Azure Synapse dedicated SQL pools because Azure Synapse can act as our high performance allowing for complex queries on large datasets without slowing down the source system.
b) **Insight Generation:** Using business-ready data, we developed interactive visualizations. Because the naming was cleaned and dates were standardized in Databricks, the reports provide clear, accurate metrics such as sales growth and customer demographics.

4.4 SECURITY AND GOVERNANCE

We managed the security and the number of users by using Azure Active Directory and Key vault. By using Azure Entra ID, we can manage the users identities and enforce Role-Based Access Control (RBAC). Thus, instead of giving everyone every permissions, we are able to assign a special role as a "Contributor" for the pipeline to avoid unauthorized users to make changes to the pipeline.

Azure Key Vault further enhances our project security with the Key Vault Encryption technology where we only save usernames and passwords in the secret of the key vault, we are able to use that information with encryption keys ensuring passwords remain hidden and secure from external leaks.

5.0 MS POWER BI DASHBOARD

The MS Power BI dashboard converts processed data into useful business insight and acts as the last visualization layer in the end-to-end data pipeline. This section has a polished dashboard design with insightful graphics and a straightforward data storytelling strategy that emphasizes important industry findings. The dashboard shows how the complete pipeline is successfully integrated by connecting directly to the modified data kept within the Azure environment. This component's main goal is to give stakeholders a useful and user-friendly interface so they can accurately and clearly analyze complex datasets.

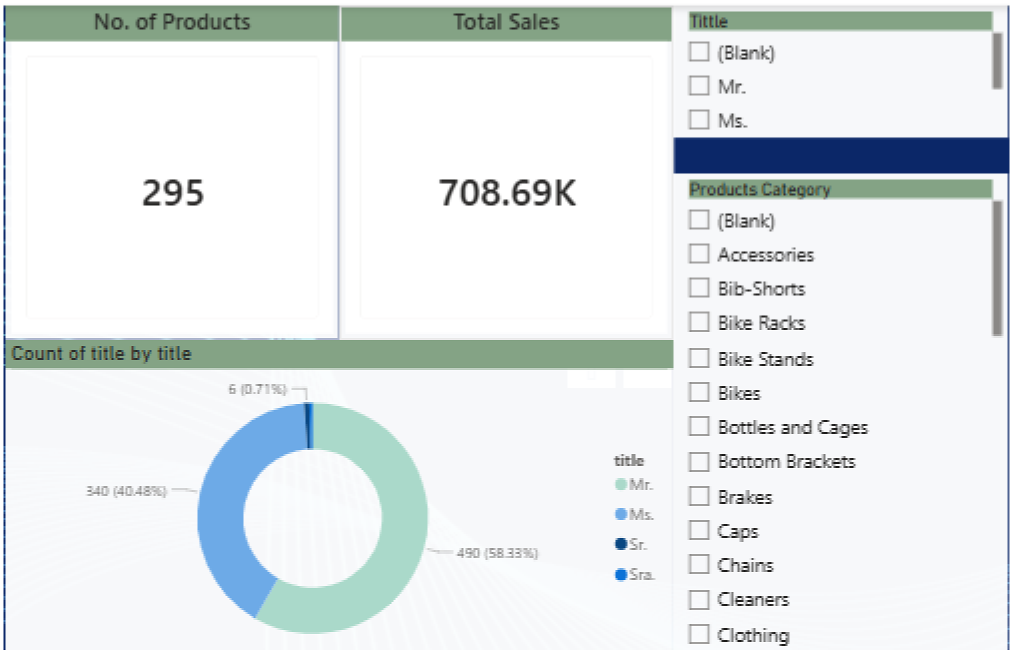

Figure 2: Power BI Dashboard

6.0 REFLECTION

6.1 NURUL ASYIKIN BINTI KHAIRUL ANUAR

This project helped me understand how a real data pipeline works, particularly how Azure services such as Data Factory, Databricks and Power BI work together to process data from beginning to end. The Bronze, Silver and Gold layering concepts were also unfamiliar to me, but after I understood the purpose of each layer, the entire data organizing process became much obvious.

I initially struggled to understand how data moves between the various systems, but going through it step by step and collaborating with my group members helped me overcome this. Overall, this project provided me with invaluable hands-on experience with Azure, which increased my confidence in data engineering.

6.2 REFLECTION BY LUBNA AL HAANI BINTI RADZUAN

Working on this tutorial helped me better understand the medallion architecture, not just in terms of structure but also its purpose. The three layers, bronze, silver, and gold, are used to separate data into different tiers based on how refined it is, which makes the data easier to manage, clean, and use for analysis.

I also gained a clearer picture of the overall workflow, including how data moves from Azure Data Factory to Microsoft Power BI. Along the way, I learned about managing user access and permissions, as well as how Azure Key Vault helps securely store sensitive information like connection strings and keys. This made me understand how both data flow and security are handled together in a real-world data engineering setup. To conclude, this hands-on tutorial made me learn the process of each stage and also its possible errors and how to fix them.

6.3 ANIS SAFIYYA BINTI JANAI

Before completing this tutorial, my concept of data pipeline is still vague. When I switched from the local SQL database to the Azure data factory, my technical perspective changed significantly. I understand the organizational structure between the conversion layer and Azure Data Lake Storage (second generation), which was one of the biggest obstacles I encountered. At first, there were some difficulties in configuring the link service and ensuring that the authentication protocol settings were correct to achieve smooth data flow. After overcoming these connection problems, I realized that security and integration are as important as data processing in the cloud architecture.

I especially want to learn more about Azure Databricks so that I can manage more complex and extensive data conversion in the future. This project is not only a technical exercise, but also a key step in understanding how modern enterprises can use professional dashboards to turn unprocessed isolated data into executable insights.